

**APRIL 2025**

**2504235 / UMSA32I**

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Time: Three Hours

Maximum Marks: 75

**SECTION - A ( $10 \times 2 = 20$  Marks)**

**Answer Any TEN Questions**

1. What are the steps involved in the execution of the plan.
2. Define statistics.
3. Write down the two popular methods of sampling.
4. Define diagrams.
5. Write down the percentage formula in bar diagram.
6. Write down the two types of graphs in identification number.
7. Define arithmetic mean.
8. Find the mode: 3 6 7 5 8 4 9.
9. Define Skewness.
10. Write down the formula for Karl Pearson's coefficient of skewness.
11. Define principle of least square.
12. Write down the spearman's rank correlation coefficient.

**SECTION - B ( $5 \times 5 = 25$  Marks)**  
**Answer Any FIVE Questions**

13. Write a short note on collection of data
14. Population of India in 5 censal years is given. Represent this by simple bar diagram.

| Year | Population(crores) |
|------|--------------------|
| 1951 | 36                 |
| 1961 | 44                 |
| 1971 | 55                 |
| 1981 | 68                 |
| 1991 | 84                 |

15. Write down the merits of median.
16. Calculate G.M for the data given below

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | 10 | 12 | 25 | 40 | 50 |
| f | 4  | 6  | 10 | 7  | 3  |

17. Calculate Karl Pearson's coefficient of skewness for the following data.
- 25    15    23    40    27    25    23    25    20

18. Calculate Bowley's coefficient of skewness.

|                               |   |    |    |    |    |    |   |
|-------------------------------|---|----|----|----|----|----|---|
| No. of of Children per family | 0 | 1  | 2  | 3  | 4  | 5  | 6 |
| No. of Families               | 7 | 10 | 16 | 25 | 18 | 11 | 8 |

19. Fit a straight line.

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| X | 2 | 3 | 5  | 8  | 10 |
| Y | 5 | 6 | 10 | 18 | 21 |

**SECTION - C ( $3 \times 10 = 30$  Marks)**  
**Answer Any THREE Questions**

20. The following data indicates the rice (in Kg's) consumption of 30 families. frame the frequency table with this data.

30 20 14 52 24 33 56 45 24 37

21 43 25 11 33 19 26 31 42 34

37 41 15 47 21 38 26 44 28 12

21. Explain in brief about general rules of graphs.
22. Calculate the S.D of the following series.

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| X | 6 | 9  | 12 | 15 | 18 |
| f | 7 | 12 | 13 | 10 | 8  |

23. Calculate A.M, S.D,  $\gamma_1$  and  $\gamma_2$  and comment upon the nature of skewness and Kurtosis.

|   |   |    |    |    |   |
|---|---|----|----|----|---|
| X | 5 | 6  | 7  | 8  | 9 |
| f | 5 | 10 | 24 | 15 | 6 |

24. Calculate the two regression equations from the following data.

|   |    |    |    |    |    |    |
|---|----|----|----|----|----|----|
| X | 10 | 12 | 13 | 12 | 16 | 15 |
| Y | 40 | 38 | 43 | 45 | 37 | 43 |

Also estimate Y when X=20.